QUAID-E-AWAM UNIVERSITY OF ENGINEERING SCIENCE & TECHNOLOGY, NAWABSHAH

BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)

Course Information		Course Assessment	
Title of Course:	**Pre-Calculus I**	**Theory:**	20% sessional work 20% mid-semester 60% final examination
Course Code:	None		
Pre-requisite :	None		
Year / Semester:	First Year, First Semester	**Practical:**	N/A
Marks (Th + Pr):	00 + 00		
Credit Hours:	3 + 0	**Tools:**	Assignments, Tests Mid-term exams, Final Exams
Contact Hours:	45 + 00		

Course Learning Outcomes

Upon completion of this course, students will be able to achieve the following objectives:

CLO's		Domain	Taxonomy level	PLO
CLO:1	**Apply** differential and integral calculus to interpret the physical systems and processes	Cognitive	3	2
CLO:2	**Identify** real and complex numbers and understand the behaviour of sequence and series	Cognitive	1	1

Course Contents:

Defining Set, various types of set representation and operations, Relation and function, Properties of functions, composition and inverses of functions, domain and range of the functions, Maximum and minimum values of functions, increasing and decreasing functions, zeros and intercept of functions, piecewise functions, continuity and Discontinuity of functions, Polynomials and rational functions, Polynomial long division and Synthetic division, Solution of rational functions, Absolute valued function, properties of absolute valued functions, Asymptotes (Horizontal, vertical and oblique), Exponential functions and their properties, Logs functions and their properties, Systems of Two Equations and Two Unknowns, Systems of Three Equations and Three Unknowns, Matrix Algebra (Add, subtract and multiply matrices), Row Operations and Row Echelon Forms, Augmented Matrices, Determinant of Matrices (2 x 2 and higher order matrices), Cramer's Rule, Inverse Matrices, Series and Sequences

Textbooks:

[1] Mathematics for Class Ninth IX published by Sindh Textbook Board Jamshoro, Pakistan
[2] Mathematics for Class Twelve XII published by Sindh Textbook Board Jamshoro, Pakistan
[3] Textbook of Algebra and Trigonometry Class XI published by Punjab Textbook Board (PTB) Lahore, Pakistan.
[4] Calculus and Analytic Geometry, MATHEMATICS 12 (Mathematics FSc Part 2 or HSSC-II), Punjab Textbook Board Lahore, Pakistan